Adaptive Feature Gating Enhanced Stable Diffusion for Large-Scale Hyperspectral Image Super-Resolution

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***Abstract*—Hyperspectral image super-resolution (HSIs SR) aims to enhance the spatial resolution of HSI data without requiring costly hardware modifications. While deep learning methods have shown success with 4× and 8× scaling, achieving 16× SR remains a significant challenge. This difficulty primarily arises from the lack of sufficient reference information and effective domain-specific priors, which hinders the accurate preservation of intricate spectral and spatial details. Recent advances in stable diffusion (SD) models, pre-trained on large-scale datasets, offer a promising solution by leveraging effective image priors. However, ControlNet-based SD frameworks often suffer from color-shift artifacts in generated images. To address this, we propose a novel Adaptive Feature Gating-Enhanced Stable Diffusion (AFG-SD) framework tailored for 16× HSIs SR. Our framework includes an Adaptive Feature Gating (AFG) mechanism that adjusts the encoder representations within the SD module, ensuring both color consistency and structural congruence with the low-resolution input. Furthermore, we utilize the spectral low-rank properties of HSI data by employing a rank K-SVD algorithm to estimate a coefficient matrix. This enables the transformation of reconstructed three-band RGB images into multi-band HSIs. Quantitative and qualitative analyses on the ICVL and CAVE dataset demonstrate that our method significantly outperforms state-of-the-art approaches across key metrics, including SSIM, LPIPS, DISTS, FID, CLIPIQA, and MUSIQ.**

***Index Terms*—16× HSIs SR, Stable Diffusion (SD) model, ControlNet, Adaptive Feature Gating, Low-rank property.**

I. Introduction

Hyperspectral imaging offers rich spectral information for a variety of applications [1], but the trade-off between spectral and spatial resolution often limits its utility. Hyperspectral image super-resolution (HSIs SR) addresses this by enhancing spatial detail through computational methods, avoiding costly hardware upgrades. While conventional methods and deep

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SDP [7] TIP 24 | ControlNet [8] | Ours | GT

Fig. 1: Illustration of the SR results of comparative approaches and ours.

learning (DL) approaches have made progress, a significant challenge remains in achieving 16x SR, which is particularly critical in applications demanding sub-meter spatial precision. For instance, while WorldView-3 offers a 0.31 m panchromatic resolution, its multispectral resolution is only 11.1 m, 16× SR enhances this to 0.69 m, and the enriched spectral data permits sub-pixel analysis vital for detecting camouflaged targets in military reconnaissance [2, 3], microscopic imaging [4], and satellite image [5, 6]. 16× SR is particularly difficult due to the lack of sufficient domain-specific priors, which can lead to artifacts and a loss of fine detail in existing methods.

Existing deep learning-based methods can be broadly classified into two categories: discriminative models and generative models. Discriminative models, like those using spatial-spectral attention modules [7], focus on refining details by capturing dependencies across bands. Generative models, such as diffusion models [9, 10], aim to learn the underlying distribution of high-resolution HSIs to create realistic outputs. However, most end-to-end DL approaches fail to incorporate additional prior knowledge effectively, leading to degraded performance in ultra-high SR tasks, as seen in the artifact-ridden results from a discriminative model shown in Fig. 1.

Recently, Stable Diffusion (SD) models [8], pretrained on massive image datasets, have shown great potential by providing rich image priors. However, a direct application of ControlNet-based SD frameworks to HSIs SR introduces a significant color-shift problem, as shown in Figure 1. To overcome this, we propose a novel Adaptive Feature Gating-Enhanced Stable Diffusion (AFG-SD) framework. Our key innovation is an Adaptive Feature Gating (AFG) mechanism that modulates the SD encoder's representations using features from the low-resolution input. This ensures that the generated images maintain color consistency and structural fidelity with their source. Additionally, we use a rank K-SVD algorithm to leverage the spectral low-rank property of HSIs, transforming the three-band RGB output of the SD model into a full-spectrum HSI. The core contributions of our work are:

- The development of an AFG-SD framework for 16x HSIs SR that effectively leverages powerful pretrained image priors.
- The introduction of an Adaptive Feature Gating (AFG) module to mitigate color-shift artifacts and ensure structural congruence.
- A coefficient matrix estimation strategy that enables the reconstruction of multi-band HSIs from a three-band output. Quantitative and qualitative analyses have demonstrated the superiority of the proposed approach.

II. The Proposed Super-resolution Model

Consider a HR HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, where H and W denote the height and width of the HSI, and B represents the number of spectral bands. The degradation process can be modeled as follows,

$$\mathcal{Y} = (\mathcal{X} * k) \downarrow_{16} + n, \tag{1}$$

where $\mathcal{Y} \in \mathbb{R}^{h\times w\times B}$ denotes the corresponding LR HSIs, and $h, w = H/16, W/16$, and n is the additive Gaussian noise.

If we directly leverages SD priors to achieve a high 16× SR, thereby generating high-fidelity images. Nevertheless, directly applying the SD model to HSIs SR introduces two main challenges. **First**, the Controlnet-based SD framework tends to generate images with color shifts. **Second**, the original T2I model is constrained to generating three-channel images, which is incompatible with the $B(B > 3)$ spectral bands inherent to HSIs. We subsequently elaborate on the methodologies employed to address these two challenges.

To address the first problem, we introduce AFG mechanism to dynamically adjust the SD encoder features. Specifically, we adopt the pre-trained T2I SD model as the foundational backbone of our framework. Subsequently, ControlNet is employed as a controller to guide the conditional image generation process. ControlNet is initialized as a trainable replica of the encoder from the U-Net architecture inherent to the SD model.

In order to extract discriminative features from LR images that effectively enhance the efficacy of the generation process, we propose an Initial Restoration Block (IRB) composed of residual blocks, 3×3 convolutional layers, and a four-stage spatial rearrangement cascade. This architecture is designed to mitigate degradation factors while progressively restoring the LR image to its native resolution through hierarchical feature refinement

$$\mathbf{F}_c = \mathrm{IRB}(\mathcal{Y}), \qquad \gamma, \beta = \mathrm{split}\big(\mathrm{NLC}(\mathbf{F}_c)\big),$$

where NLC denotes a non-linear control module, implemented as two successive convolutional layers followed by Leaky ReLU activations, while split$(\cdot)$ partitions the resulting tensor along the channel dimension.

To correct the global luminance and chromatic mismatch often observed between the SD output and the LR reference, we modulate the encoder features in the frequency domain. For the k-th feature map x_k,

$$\widehat{x}_k = \mathcal{F}\{x_k\}, \quad \widehat{x}_k^{\mathrm{m}}(1+\gamma)\odot\widehat{x}_k+\beta, \quad x_k^{\mathrm{m}} = \mathcal{F}^{-1}\{\widehat{x}_k^{\mathrm{m}}\},$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and its inverse, respectively. We first extract three primary spectral bands, specifically the 22nd, 14th, and 7th bands, from the HR HSI and acquire corresponding RGB images x_{hr}, and then encode them into a latent embedding z_{hr}. The training process is divided into two stages, in the first stage, we fine-tune the VAE within the ICVL Dataset by adopting the following training objective,

$$\mathcal{L}_{\mathrm{vae}} = \|\mathcal{D}(\mathcal{E}(x_{hr})) - x_{hr}\|_1 + \omega \cdot \mathrm{KL}\left(N\left(u, \sigma^2\right) \| N(0,1)\right), \tag{2}$$

where w denotes the regularization parameters, and μ, σ indicate the mean and variance of the reconstruction distribution, respectively. During the two stage, the latent representation is perturbed by Gaussian noise, and at time step t, it is represented as $z_{hr}^t = \sqrt{\bar{\alpha}_t} z_{hr} + \sqrt{1-\bar{\alpha}_t}\epsilon$. The proposed model aims to predict the noise ϵ added to the latent representation z_{hr}. This prediction is conditioned on the LR images x_{lr} corresponding to the three main spectral bands of $\mathcal{Y}$, the current time step t, and an accompanying text prompt c. Formally, the optimization objective is expressed as,

$$\begin{aligned}\mathcal{L} = \mathbb{E}_{z_{hr},t,c,x_{lr},\epsilon\sim\mathcal{N}(0,1)} &\left[\left\|\epsilon - \epsilon_\theta\left(z_{hr}^t, t, c, x_{lr}\right)\right\|_2^2\right] \\ &+ \|x_{hr} - IRB(x_{lr})\|_F^2\end{aligned} \tag{3}$$

To address the second problem, in order to produce HSIs with B ($B > 3$) spectral bands. Due to the spectral low-rank property inherent in HSI data, we can utilize the rank K-SVD algorithm to estimate the coefficient matrix $\mathbf{E}$, let $\mathbf{Y}_{(3)} \in \mathbb{R}^{B\times hw}$ denotes the mode-3 unfolding of $\mathcal{Y}$,

$$\mathbf{Y}_{(3)}^{\mathrm{T}} = (\mathbf{US})\mathbf{V}^{\mathrm{T}}, \tag{4}$$

where $\mathbf{U} \in \mathbb{R}^{hw\times B}$, $\mathbf{S} \in \mathbb{R}^{B\times B}$ and $\mathbf{V} \in \mathbb{R}^{B\times K}$. Denote $\mathbf{V}_s = \left[\mathbf{v}_{i_1}^{\mathrm{T}}, \mathbf{v}_{i_2}^{\mathrm{T}}, \cdots, \mathbf{v}_{i_K}^{\mathrm{T}}\right]^{\mathrm{T}}$, where $\mathbf{v}_{i n}$ represents the i_n-th row of $\mathbf{V}$. According to [13, 9], the coefficient matrix can be estimated $\mathbf{E} = \mathbf{V}\mathbf{V}_s^{-1}$. Here, we set $K = 3$ for constructing the coefficient matrix $\mathbf{E} \in \mathbb{R}^{B\times 3}$ to align with the three-channel images generated by T2I diffusion model and multi-spectral band HSIs. The recovered HSIs can be estimated as $\hat{\mathcal{X}} = \mathcal{A} \times_3 \mathbf{E}$.

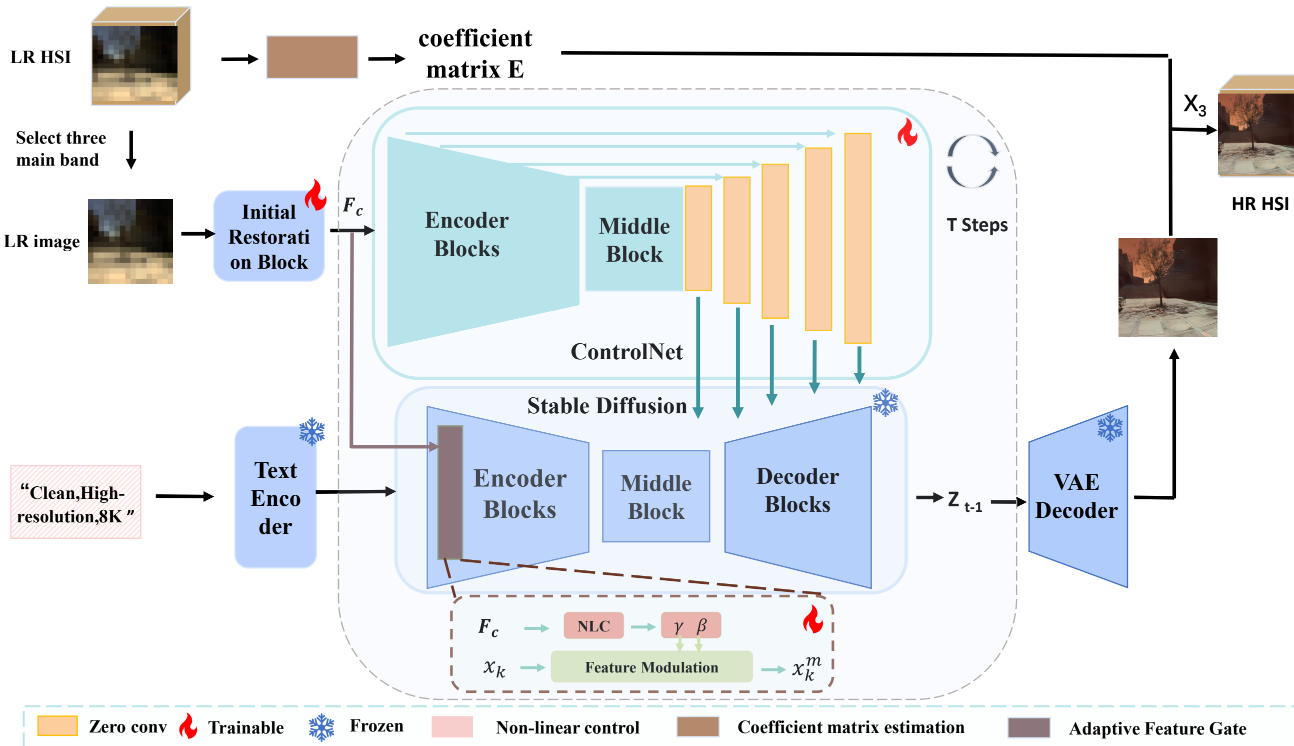

Fig. 2: The overall framework of our proposed 16× HSIs SR framework. First, the coefficient matrix **E** is estimated using the rank-SVD approach. Subsequently, the LR image is input into ControlNet, which functions as a controller to guide the diffusion process in generating a HR image. Finally, the restored HSIs is obtained through the tensor product of the estimated coefficient matrix **E** and the recovered images $\mathcal{A}$.

TABLE I: Quantitative comparison with SOTA methods on ICVL and CAVE datasets. Red and blue colors indicate the best and second-best performances, respectively. Results are reported as **ICVL / CAVE**.

Dataset	Metrics	Wen *et al.* [11]$^{\text{TCSVT 21}}$	SelfDeblur [12]$^{\text{CVPR 20}}$	HIR-Diff [9]$^{\text{CVPR 24}}$	SDP [7]$^{\text{TIP 24}}$	Ours
ICVL / CAVE	PSNR ↑	20.30 / 16.89	16.75 / 14.92	18.57 / 16.13	25.90 / 21.30	23.91 / 20.90
	SSIM ↑	0.6199 / 0.6906	0.4159 / 0.4100	0.5983 / 0.6059	0.7119 / 0.7339	0.8549 / 0.8021
	LPIPS ↓	0.5129 / 0.5956	0.6043 / 0.8012	0.6345 / 0.6887	0.6361 / 0.4992	0.3180 / 0.3672
	DISTS ↓	0.3818 / 0.3874	0.3706 / 0.5117	0.4400 / 0.4572	0.4054 / 0.3843	0.2427 / 0.2943
	FID ↓	353.90 / 331.80	366.21 / 378.08	362.67 / 310.73	318.29 / 295.30	136.08 / 184.72
	CLIPIQA ↑	0.1368 / 0.1397	0.2583 / 0.3025	0.2128 / 0.1377	0.1762 / 0.1476	0.3468 / 0.3282
	MUSIQ ↑	29.39 / 24.37	21.40 / 25.77	17.89 / 22.21	17.00 / 13.95	39.31 / 26.21
	SAM ↓	8.4231 / 8.5612	12.1542 / 12.8943	9.8765 / 10.1245	6.1248 / 6.4531	4.5621 / 4.8722
	Time (s) ↓	272	166	193	1.07	16.32 / 1.06*

* denotes one-step diffusion through DMD distillation.

III. Experimental Results and Discussions

A. *Experimental Settings*

Training datasets. We evaluate the proposed approach using the ICVL dataset [14] and CAVE Dataset[15]. Initially, all HSIs are cropped to a resolution of 256 × 256 × 31 pixels. Utilizing the degradation pipeline described in Equation (1), we construct more than 2000 LR and HR HSIs pairs by applying different convolution operators. The dataset is subsequently partitioned into training, validation, and test sets, comprising 90%, 5%, and 5% of the data, respectively.

Implementation Details. We employ the pretrained SD v1.5 model as the foundational T2I framework. During the training phase, the AdamW optimizer is utilized.

Evaluation Metrics. We utilize a variety of commonly used metrics to evaluate the effectiveness of proposed approach, encompassing both reference-based and non-reference metrics. The former comprises of PSNR, SSIM [16], LPIPS [17], and DISTS [18] and FID [19]. Among them, PSNR and SSIM are employed to quantify discrepancies at the pixel level, whereas FID, DISTS, and LPIPS assess image quality from the perspective of human visual perception. In contrast, CLIPIQA [20], and MUSIQ [21] serve as non-reference metrics for evaluating image quality.

B. *Experimental Results*

To validate the effectiveness of proposed approach, we compare the proposed approach with other SOTA HSIs SR

Fig. 3: 16× HSIs SR results for two scenarios via different comparative approaches. Apart from PSNR, the proposed method attains the best scores on every quantitative metric and produces reconstructions that are perceptually more realistic and contain finer spatial detail than those generated by the comparative approaches.

methods, including Wen *et al.* [11], SelfDeblur [12] , HIRDiff [9], and SDP [7].

Quantitative Comparison. Table I presents the quantitative results of the proposed method compared to other state-of-the-art approaches on the ICVL dataset. **First**, the proposed approach outperforms all competing methods in the SSIM, LPIPS, DISTS, FID, CLIPIQA, and MUSIQ metrics by substantial margins, exceeding the second-best method by 20.01%, 37.99%, 37.55%, 57.24%, 34.26%, and 33.75%, respectively. **Second**, the proposed approach falls behind end-to-end DL methods in metrics such as PSNR. This is because the diffusion-based approach prioritizes perceptual quality and structural details over exact pixel-wise accuracy, resulting in lower PSNR scores. Additionally, the visualization results in Fig. 3 indicate that **the proposed approach generates higher-quality images in terms of human perception, despite underperforming in some pixel-level reference metrics.** This phenomenon has also been explored in recent studies [22, 23].

Vision comparison. Fig. 3 present qualitative comparisons on the test sets. Other comparative approaches fail to recover images with acceptable quality due to the 16× downsampling operator, which results in reduced reference information. In contrast, the proposed method leverages the effective priors encapsulated in the pretrained diffusion model to generate more realistic images. Additionally, the AFG mechanism contribute to mitigate the color shift problem, yielding generated images that closely align with the corresponding GT images.

C. Ablation Study and Discussions

TABLE II: Ablation Study of Key Components for 16× HSIs SR.

Method	**PSNR ↑**	**SSIM ↑**
Ours w/o AFG	20.45	0.7218
Ours w/o VAE fine-tuning	18.92	0.6833
Ours	**23.91**	**0.8549**

Effectiveness of the proposed adaptive feature gating mechanism. In this section, we aim to evaluate the important role of proposed AFG mechanism in preserving the color and feature consistency with input LR images. To this end, we remove the AFG and perform 16× SR while keeping all other architectures fixed. The results of the experiments are shown in Table II, show a noticeable decline in PSNR and SSIM metrics when discarding the AFG mechanism. These results demonstrate that dynamic re-weighting of encoder features performed by the AFG module is pivotal for generating HSIs that are both colour consistent and structurally faithful to the LR observations.

Effectiveness of fine-tuning VAE in SD. To evaluate the impact of VAE fine-tuning in the SD framework, we performed an ablation study that re-uses the pre-trained VAE without additional fine-tuning, while keeping all other network components unchanged. The quantitative results, summarized in Table II, reveal clear improvements in both PSNR and SSIM after fine-tuning. These gains indicate that incorporating domain specific knowledge substantially enhances the VAE's capacity to produce restorations with more consistent and faithful color.

IV. Conclusion

In this work, we developed a AFG-SD framework for 16× HSIs SR. The proposed approach introduces an AFG mechanism that refines SD encoder feature representations to mitigate the color shift problem. A coefficient matrix approach is employed to align with the three-channel images generated by T2I diffusion model and multi-spectral band HSIs. Consequently, our method not only support ultra-high scaling factors but also delivers superior performance in preserving the intricate details essential for accurate HSIs analysis. Future work will establish more suitable optimization conditions to minimize the pixel-level divergence between generated and ground truth images.

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